



BACHELOR OF INFORMATION TECHNOLOGY (HONS)

FINAL EXAMINATION OCTOBER 2024

Course: EC3130 (Introduction to Data
Communication and Networking)

Time: 8:00 am – 11:00am (3 hours)

Lecturer: Mr. SEOW Soon Loong

Date: 09 January 2025

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of 3 printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO SO

1. a. For each of the following network topologies: Star, Ring, and Bus (18 MARKS)
- Provide a brief description of how the topology works.
 - Identify **TWO (2)** advantages and **TWO (2)** disadvantages of each topology.
- b. In the context of the TCP/IP model, demonstrate your understanding (7 MARKS)
by matching each layer of the OSI model with its corresponding layer(s) in the TCP/IP model. Provide a brief explanation for each match.
2. a. Copper cabling is widely used in modern networks due to its (8 MARKS)
availability and affordability. Identify **TWO (2)** limitations of copper cabling and explain how each limitation can be mitigated.
- b. Explain how the Carrier-Sense Multiple Access with Collision (5 MARKS)
Detection (CSMA/CD) protocol handles a collision when it occurs during data transmission.
- c. Provide a clear explanation of how the CSMA/CA protocol works (12 MARKS)
and outline the process followed by a station when attempting to transmit data.
3. a. A signal has a period of 500 microseconds (μs). You are tasked with (3 MARKS)
designing a system that requires the signal's frequency to be calculated in kilohertz.
- b. A 20-MHz bandwidth analog audio signal is digitized to create a (4 MARKS)
digital audio channel. The signal is sampled at twice the highest frequency (two samples per hertz). Assume each sample requires 10 bits. What is the required bit rate for this digitized audio signal?
- c. Given the binary data stream "011000", explain how the Polar (6 MARKS)
Return-to-Zero (RZ) encoding scheme would represent this data. Provide a detailed illustration of your explanation with a diagram.
- d. Compare and contrast the Manchester and Differential Manchester (12 MARKS)
encoding schemes in terms of signal transitions. Provide an example of how each scheme would behave with a data sequence of "110110". Illustrate your answer with diagrams.

4. a. You are tasked with designing a company's IPv4 network using Classless Addressing. The network consists of the following components: (12 MARKS)

- Four Local Area Networks (LANs) for different departments:
 - Administration (16 hosts)
 - Human Resources (6 hosts)
 - Marketing (30 hosts)
 - Operations Management (48 hosts)
- Two Wide Area Networks (WANs), WLink1 and WLink2, to support interconnectivity between sites.

The company has been allocated the Class C IP address 192.168.3.0/24.

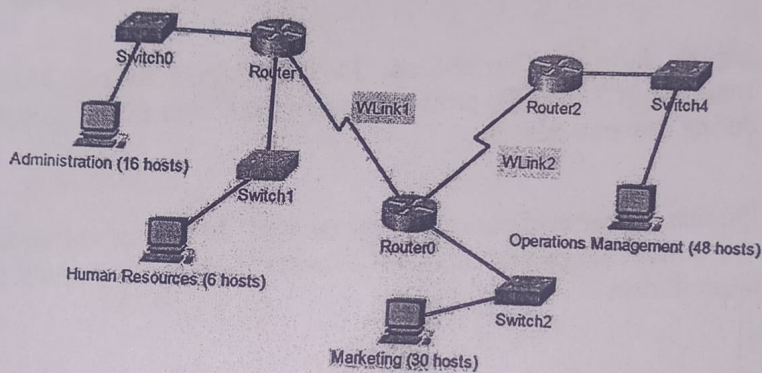


Figure 4.1

For each subnet, clearly show the following:

- Subnet address
- Subnet mask
- Range of usable IP addresses
- Broadcast address

Provide your answer using the format given in table 4.1

Subnet Name	Subnet address	First usable host address	Last usable host address	Broadcast address	Subnet Mask	Number of addresses
Operations Management	192.168.3.0/26					
Marketing	192.168.3.64/27					
Administration	192.168.3.96/27					
Human Resources	192.168.3.128/29					
WLink1	192.168.3.136/30					
WLink2	192.168.3.140/30					

Table 4.1

- b. Provide a step-by-step example of shortening the following IPv6 address using the rules for IPv6 address compression: 2001:0db8:00:000:0000:6ale:0260:8221 (3 MARKS)
- c. Consider a data frame represented by the binary sequence 100100 and a CRC polynomial divisor of 1101. Perform the binary division of the data frame by the divisor to compute the CRC value. Show all steps of the division process, including the final CRC remainder. Assume the divisor is a 4-bit polynomial and that no padding is added before the division process. (10 MARKS)

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